Eigen-Séta white paper: study report

Evidence for `crypto:papers/eigen-seta-whitepaper/`. Measured on a shared 4-vCPU Xeon @ 2.10 GHz VM (L0); security levels derived.

Timings (all cores, medians)

set	keygen	encaps	decaps	pk = ct	sk
L1	4.6 s	0.20 s	1.20 s	1344 B	1855 B
L3	84 s	1.1 s	5.8 s	1992 B	2730 B
L5	51 s	2.3 s	12.8 s	2632 B	3594 B
Q3	14 min (4 jobs)	5.9 s	37 s	3904 B	5311 B
Q5	33 min (4 jobs, 1 key)	13.7 s	89 s	5192 B	7050 B

Run run-20261017T2022Z-kem-bench, eigenx `d89d790`; every result checked. The attacker's chain step costs 618–670 F_{p^2} multiplications per step, against the cost model's 256.

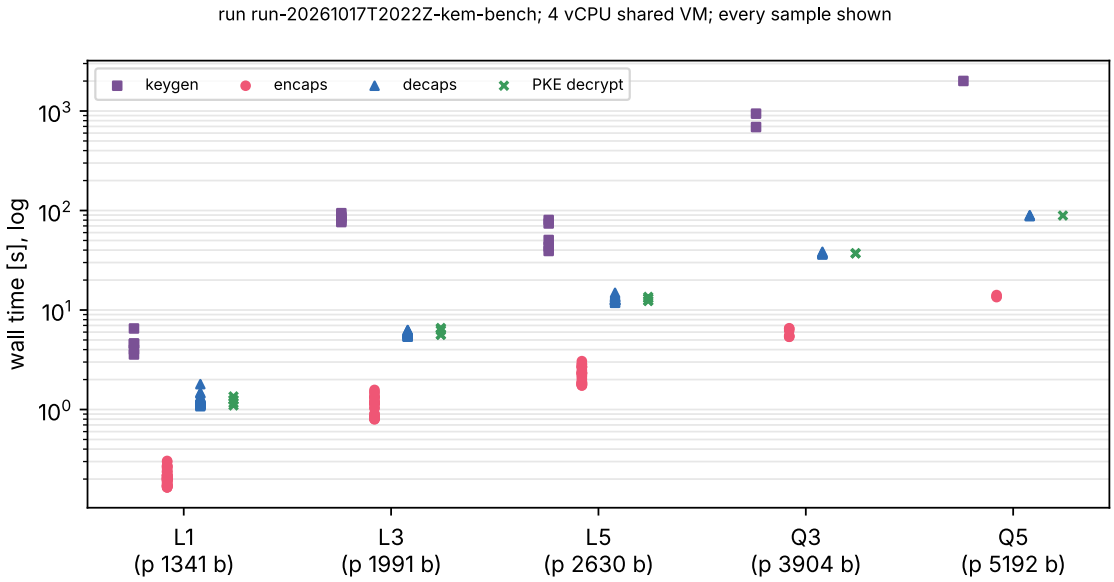


Figure 1: Every sample, log scale; measured.

Same-host comparison

Decapsulation at level 1: ML-KEM-512 14 µs, McEliece-348864 158 µs, FrodoKEM-640 441 µs, BIKE-L1 458 µs, HQC-128 (portable) 5.1 ms, Eigen-Séta L1 1.2 s, about 89,000 times ML-KEM. liboqs 0.13.0, one pinned core, two repetitions.

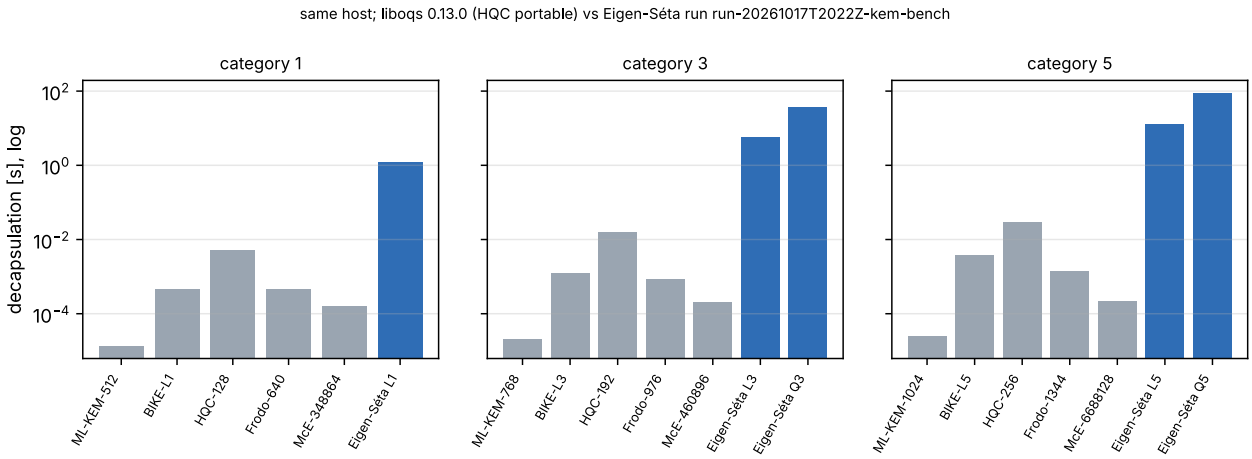


Figure 2: Decapsulation by category, log scale; measured.

Security (derived)

L1, L3 and L5 meet categories 1, 3 and 5 only in gate count under MAXDEPTH. At least 2^λ quantum queries holds only for L5 (λ = 128), Q3 and Q5. Largest risk: the hidden orientation becoming effective, which would break every set quantumly.

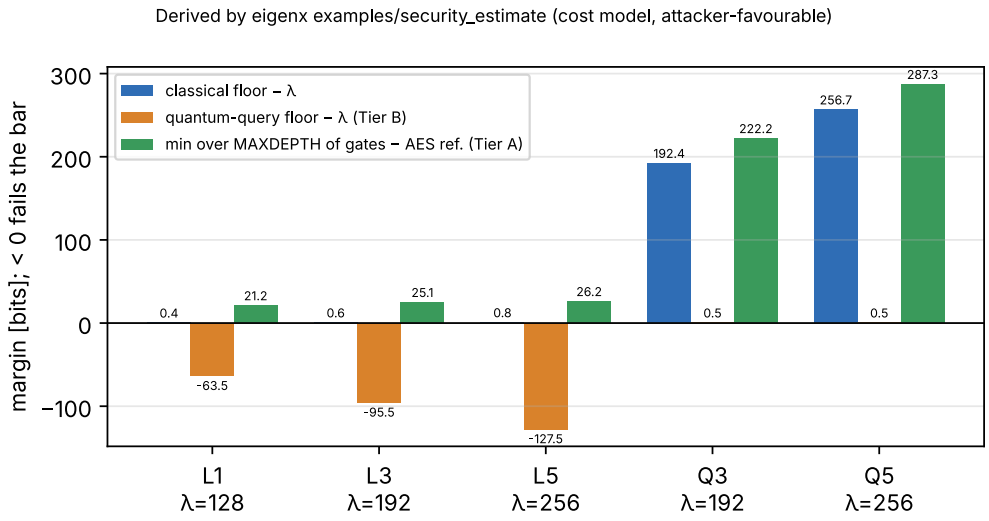


Figure 3: Margins over λ by tier; derived by the eigenx cost model.